Phase
1
2

3

Task
1. Project plan
Complete and submit a Project Plan
2. Set up Raspberry Pi as Server and confirmed communication
Set up Raspberry Pi as a server
Confirm communication between client computer and server (HTTP/SSH)
3. STM32 Programming via HAL
Explain common CAN messaging protocol (see document)
Implement the CAN messaging protocol by properly setting the filters (code demo) on STM32
Implement Wiring on STM32, push buttons/LED indicators (possible COVID restrictions)
Able to read CAN messages placed on the CAN bus (Raspberry Pi) and detect up/down request, flo request, door open/close, location of elevator car
Develop and implement a test plan to confirm functionality of software/hardware
3. Controls / indicators (virtual and buttons)
Controls (up/down) built in HTML (possibly add CSS if students work ahead)
Elevator Car Controls (up/down/floor number) built in HTML (possibly CSS if students work ahead)
4. Finite State Machine (recommended)
Develop Finite State Machine Logic for Elevator Operation
1. User Interface / Client Program
Fully working and completed front end Graphical User Interface /Webpage to control elevator
User action buttons/icons implemented
Indicators/lights etc. implemented
User action buttons developed to make changes to UI that are reflected in the database
User indicator buttons change - changes in database are reflected in UI
2. Database Repository
Database set up and accepting values/commands from website
Database has fields for logging information placed on the CAN bus
3. Diagnostic Reports
Produced based on content in database
Displayed on webpage
4. Server Program
Users are authorized/authenticated based on username/password
Server program provides logging data/accepts commands only from authorized users
5. Announce floor numbers (Function of the Client or Server Program) for users with Visual Impairmen

Implement a Server Program for announcing the floor numbers
Speaker on elevator used to announce the current/next floor (speaker already connected to syster
6. Sabbath mode (Function of the Client Program)
Users may operate the elevator without having to push buttons (e.g. verbally, based on schedule/t
1. CAN/LAN network communication
Research and Implement a CAN/LAN communication bridge
Ensure activity in CAN network changes values in database
Changes in User Interface/Website changes value in database and generate CAN messages
CAN Messages result in changes to database and GUI/Indicators
Commands from CAN/LAN networks are sent to the controller and move the elevator / indicate floo
2. Controller Software Written
CAN Supervisor (Raspberry Pi) relays messages between CAN/LAN Network (see sample program o
3. Finite State Machine logic written
Develop and Implement a finte state machine (using C/C+, PHP, Python to implement logic)
4. Additional capability that shows ingenuity
5. Ideas for Capstone (at least 3 per student)
6. Final Presentation & Demonstration

Week									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
ALL									
Fergus									
Fergus									
Nigel									
	Nigel								
	Nigel								
	Ryan								
ALL									
Fergus/Ryan									
Fergus/Ryan									
Ryan									
	Fergus								
	Fergus								
		Fergus/Ryan							
	Fergus/Ryan								
					Fergus/Ryan				
					Ryan				
					Ryan				
		Ryan/Nigel							
		Nigel							
				Fergus					
					Fergus				

						Nigel			
						Nigel			
						Ryan			
		Nigel							
		ALL							

11	12	13

*Project ends on Monday of week 14

Nigel/Ryan		
Nigel/Ryan		
Fergus/Ryan		
ALL		
ALL		
Nigel/Ryan		
Nigel		
ALL		